FIITJEE

ALL INDIA TEST SERIES

FULL TEST – III

JEE (Advanced)-2021

PAPER –1

TEST DATE: 18-01-2021

Time Allotted: 3 Hours **Maximum Marks: 198**

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-C.**

  Section-A (01 – 06, 19 – 24, 37– 42) contains **18 multiple** choice questions which have **ONLY ONE CORRECT ANSWER**. Each question carries **+3 marks** for correct answer and **–1 mark** for wrong answer.

  Section-A (07 – 12, 25 – 30, 43 – 48) *this section contains* ***18 multiple*** *choice questions.*
  Each question has FOUR options. ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct answer(s).*
  For each question, choose the option(s) corresponding to (all) the correct answer(s)
  Answer to each question will be evaluated according to the following marking scheme:
  Full Marks **: +4** If only (all) the correct option(s) is (are) chosen:
  Partial Marks **: +3** If all the four options are correct but ONLY three options are chosen;
  Partial Marks **: +2** If three or more options are correct but ONLY two options are chosen and both of which are correct;
  Partial Marks **: +1** If two or more options are correct but ONLY one option is chosen and it is a correct option;
  Zero Marks **: 0** If none of the options is chosen (i. e. the question is unanswered);
  Negative Marks **: –2** In all other cases.

  Section-C (13 – 18, 31– 36, 49 – 54) *contains* ***18 Numerical*** *answer type questions with answer XXXXX.XX and each question carries* ***+4 marks*** *for correct answer and* ***0 marks*** *for wrong answer.*

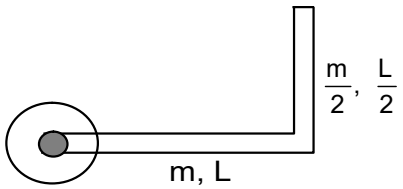
Physics

PART – I

SECTION – A
(One Options Correct Type)

This section contains **06 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. A uniform rod in the shape of L is kept in vertical plane and is hinged about an axis passing through one end what is maximum angular acceleration of rod in subsequent motion if it is released from position shown



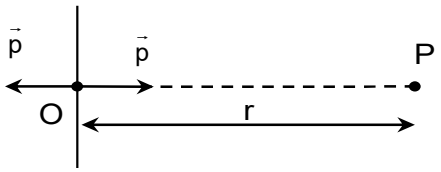
(A) $2\sqrt{65}$ g/L

(B) $\frac{\sqrt{65}}{7}.\frac{g}{L}$

(C) $\frac{8}{7}\frac{g}{L}$

(D) $\frac{16g}{9L}$

2. Two dipoles of dipole moment 'p' are kept antiparallel as shown at the origin. Length of each dipole is 'a'. Electric field at point P is



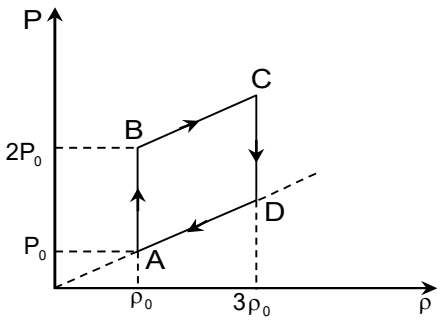
(A) $\frac{6ap}{4\pi \in_0 r^4}$

(B) $\frac{3ap}{4\pi \in_0 r^4}$

(C) $\frac{3ap}{\pi \in_0 r^4}$

(D) None of these

3. n moles of an ideal gas is taken through a cyclic process ABCD. Pressure – density diagram of which is shown in adjacent figure. If molar mass of gas is M, what will be the work done by gas in cycle.



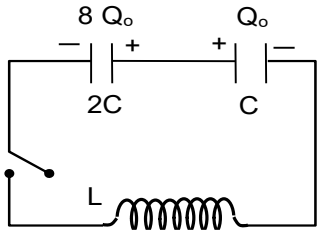
(A) $\frac{nMP_0}{\rho_0}\ell n3$

(B) $\frac{2nMP_0}{\rho_0}\ell n2$

(C) $-\frac{nMP_0}{\rho_0}\ell n3$

(D) $\frac{-2nMP_0}{\rho_0}\ell n2$

4. Two capacitors of capacitance 2 C and C are given charges 8 Q. and Q. respectively as shown in figure. If switch is closed what will be maximum current through inductor.



(A) $\sqrt{\frac{3}{LC}}Q_0$

(B) $\sqrt{\frac{6}{LC}}.Q_0$

(C) $\sqrt{\frac{2}{LC}}.Q_0$

(D) None of these

5. A plane longitudinal wave of angular frequency 1000 Hz is travelling along positive x – direction in a homogeneous gaseous medium of density $P = 1\,kg/m^3$. Intensity of wave is I = 10^{-10} w/m^2 and maximum pressure change is $(\Delta P)_{max} = 2\times10^{-4} N/m^2$. Assuming at x = 0 initial phase of medium particles to be zero, choose the equation of the wave.

(A) $\Psi = 10^{-9}\sin(1000t + 5x)m$

(B) $\Psi = 10^{-9}\sin(1000t - 5x)m$

(C) $\Psi = 10^{-8}\sin(1000t - 5x)m$

(D) $\Psi = 10^{-7}\sin(1000t - 5x)m$

6. A perfectly reflecting ball of radius R and mas m is kept on a rough horizontal floor. A parallel beam of intensity I is incident on the ball at an angle $\theta = 37^\circ$ with the horizontal. The initial acceleration of the ball is

(A) $\frac{4\pi R^2 I}{7MC}$

(B) $\frac{3\pi R^2 I}{7MC}$

(C) $\frac{5\pi R^2 I}{7MC}$

(D) None of these

(One or More than one correct type)

This section contains **06** multiple choice questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

7. Which of the following have the dimension of energy?

(A) $\frac{1}{2}\rho v^2$

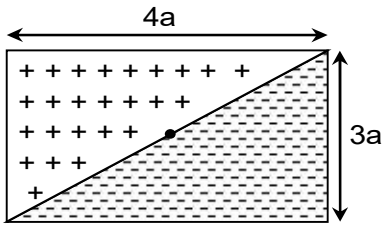
(B) $\frac{1}{2}\in_0 E^2$

(C) $\frac{B^2}{2\mu_0}$

(D) None of these

8. Choose the correct statements:
   (A) If an electron is accelerated from rest through a potential different 'V' its de Broglie wavelength becomes $\lambda = \frac{h}{\sqrt{2meV}}$
   (B) photoelectrons are emitted from metal surface when wavelength is above threshold wavelength.
   (C) stationary orbits are those orbits in which electron waves form standing wave.
   (D) X-rays have the highest penetration ability in all electronic spectrum.

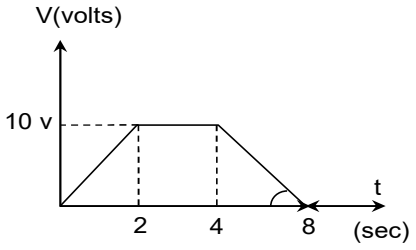
9. A non conducting rectangular plate of M is given two charges +Q and -Q in two halves which is also distributed uniformly. This plate is free to rotate in plane about an axis passing through its centre. This plate is put in uniform electric field. $\vec{E}$, and it is in equilibrium.



   (A) Dipole moment of plate is $5Qa/3$
   (B) Time period of small oscillation if disturbed from mean position is $\sqrt{\frac{5ma}{QE}}$
   (C) Torque acting on plane when it is rotated through 37° is QaE
   (D) change in potential energy when it is rotated through 53° is QaE

10. For a hydrogen like atom of atomic number z. If E, u and r reconsents energy, speed and radius of nth level choose the correct option(s).
   (A) E/v^2 is independent of n
   (B) E/Q^2 is independent of z
   (C) v^2r is independent of n
   (D) vr is independent of z

11. The potential difference a cross a 10 H inductor as a function of time is a shown in figure. At t = 0, current in inductor is zero. Then choose the correct option(s).
   (A) The current in inductor at t = 1 sec is 1/4 amp.
   (B) The current in inductor at t = 3 sec is 2 amp.
   (C) The current in inductor at t = 5 sec is 31/8 amp.
   (D) The current in inductor at t = 7 sec is 39/8 amp.



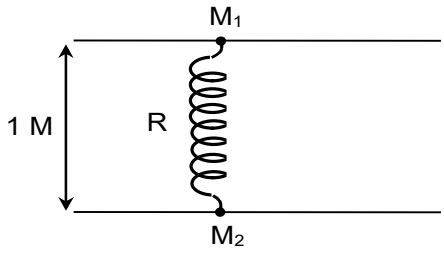
12. When a yellow light of wavelength $6000\,\mathring{A}$ falls on a plate of a photocell, a negative potential of 0.3 volts is required to stop the photocurrent completely. Then choose the correct option(s).
   (A) Work function of metal plate is nearly 2.5 V
   (B) Work function of metal plate is nearly 1.8 V
   (C) If $\lambda = 4000\,\mathring{A}$ stopping potential will be 0.6 volt
   (D) If $\lambda = 4000\,\mathring{A}$ stopping potential will be 1.3 volt.

SECTION – C
(Numerical Answer Type)

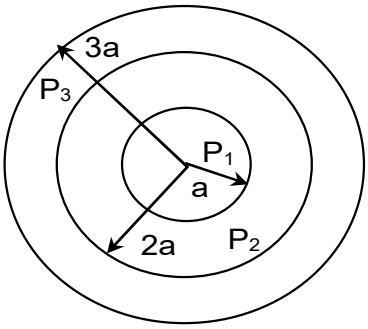
This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

13. A radioactive sample contains two type of nuclei x and y. x has half life of 5 days where as y has of 30 days. Initial activity of x is 64 times the activity of y. after how many days their activities will be equal?

14. Consider two rings of masses M_1 = 100 gm and M_2 = 200 gm connected by a spring of force constant 240 N/M and natural length 100 cm. Rings are passing through two smooth rails as shown. A ring of mass M_1 is thrown along rails with speed 12 M/s. Calculate maximum elongation of spring (in cm) during subsequent motion.

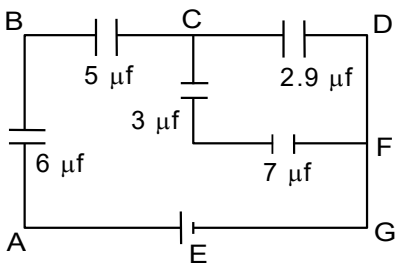


15. Consider a hypothetical planet which is made in 3 layers of different densities as shown in figure. Acceleration due to gravity at each boundary is equal and has value 'go'. If acceleration to gravity due to outer most layers just outside the planet is 'g'. Find g_0 / g.



16. Three waves $\Psi_1 = A\sin[Kx - \omega t), \Psi_2 = 3\sqrt{2}A\sin(kx - \omega t + \phi)$ and $\Psi_3 = 4A\sin(kx - \omega t + \pi/2)$ are superimposed. If resultant wave is given by $\Psi = R.\sin(kx - \omega t + \pi/2)$. Where R is some constant. Find ϕ (in degrees).

17. Consider the circuit shown. If in steady state $V_A - V_c = 11$ volt. Find $V_E - V_G$, in steady state.



18. ρ is the minimum density of required by a star such that if it rotates with time period of 1 sec, it is able to hold its mass. If $\rho = x \times 10^9$ kg/m^3. Find x.

Chemistry

PART – II

SECTION – A
(One Options Correct Type)

This section contains **06 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

19. The proposed mechanism for the reaction, $2A + B + C \rightarrow 2D$, is as follows:

$A + B \underset{k_2}{\overset{k_1}{\rightleftharpoons}} X(\text{fast})$

$X + A \xrightarrow{k_3} Y(\text{slow})$

$Y + C \xrightarrow{k_4} 2D(\text{fast})$

The rate law for the above reaction is

(A) $\frac{k_1 k_4}{k_2}[A]^2[B][C]$

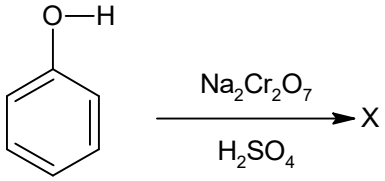
(B) $\frac{k_1}{k_2}[A]^2[B]$

(C) $k_3[A]^2[B]$

(D) $\frac{k_1 k_3}{k_2}[A]^2[B]$

20. Select the correct statement:
(A) Allene have molecular plane as nodal plane for its $\pi-$electron cloud?
(B) C_3O_2 is a nonlinear molecule
(C) Seven resonating structures are formed by tropylium cation
(D) E2 reaction always follow saytzeff

21. Which of the following is correct match?
(A) Temperature : state function : Extensive property
(B) Pressure : State function : Extensive property
(C) Internal energy : State function : Extensive property
(D) Volume : State function : Intensive property

22. 

Total number of sp^2 hybridized atom in x is,
(A) 4
(B) 6
(C) 8
(D) 7

23. $Na_2Cr_2O_7 \xrightarrow[H_2SO_4]{NaCl} \underset{(\text{Contains chromium})}{X}$

$n-\text{heptane} \xrightarrow[773\ K,\ 10-20\ atm]{Cr_2O_3} Y$

$Y \xrightarrow[ii)H_3O^+]{i)X} Z$

Number of oxygen atom in Z is ?
(A) 1
(B) 2
(C) 3
(D) 4

24. Select incorrect statement from the following:
(A) $Be(OH)_2$ is amphoteric in nature.
(B) Potassium gives violet colour in flame test
(C) The blue colour of the solution of alkali metal in liquid ammonia is due to ammoniated cation
(D) Calcium gives brick red colour in flame test

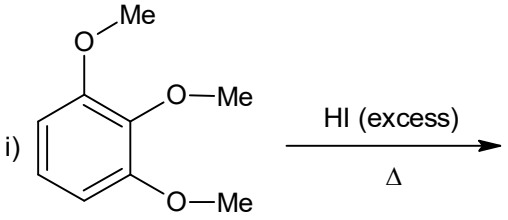
(One or More than one correct type)

This section contains **06** multiple choice questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

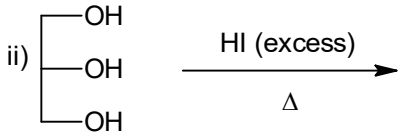
25. Select the incorrect statement:
(A) $CH_3I < CH_3Br < CH_3F < CH_3Cl$ (increasing order of dipole moment)
(B) $Si < P < S$ (increasing order of ionisation energy)
(C) $I_2 < Br_2 < F_2 < Cl_2$ (increasing order of bond energy)
(D) The number of lone pair in BF_3 & IF_2^- is same

26. Considering Bohr's model, which of the following observation is correct for one Be^{3+} ion in excited state (n = 4)?
(A) The shortest emitted wavelength is 1/15 R
(B) Six spectral lines will be seen in emission spectra
(C) The ionization energy is 2.176×10^{-18} joule
(D) The longest absorbed wavelength is $\frac{25}{9R}$

27. Consider the following reactions and pick the correct statements.

i) 

(1 mole)

ii) 

(1 mole)

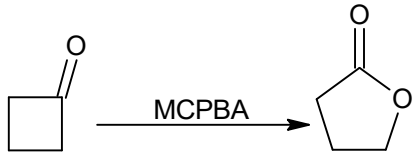
iii) 2 - butene $\xrightarrow{HI}$
(1 mole)

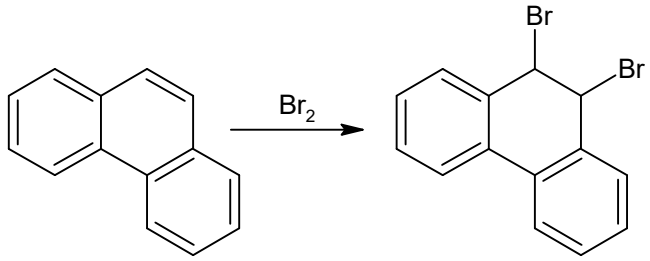
(A) More than four mole of HI is required in (ii)
(B) More than three mole of HI is required in (i)
(C) One mole of HI is required in (iii)
(D) Methanol is produced is (i), only.

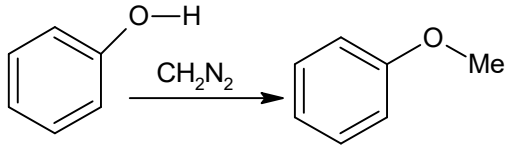
28. Select the correct statement(s):
(A) A gas with higher critical temperature gas adsorbed to greater extent than a gas with lower critical temperature
(B) $\left[Fe(CN)_6\right]^{4-}$ has higher coagulating power than NO_3^-
(C) Smaller the gold number higher the protective power of protective colloid
(D) For a solution its molarity is always having smaller value than its molality

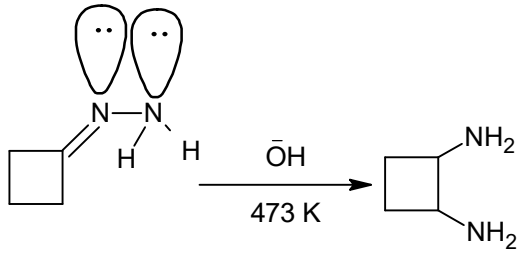
29. Butylene diamine (bn) is a chiral bidentate ligand. Now, consider $\left[Pt(bn)_2\right]^{2+}$ and pick the correct statement(s).
(A) Total number of stereoisomer is 5
(B) Two enantiomer pair
(C) Three meso compound
(D) All above statements are incorrect

30. Which of the following reaction is/are correct?

(A) 

(B) 

(C) 

(D) 

SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

31. If the EMF of the following cell is $x \times 10^{-1}$. at 25° C, then find the value of x.

$Ag \mid AgBr(s) KBr \parallel AgCl, KCl \mid Ag$

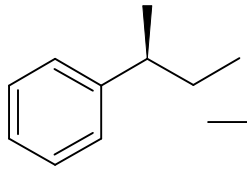
$5 \times 10^{-3} M \quad 1 \times 10^{-1}$ M

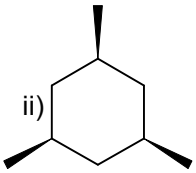
Given $K_{spAgCl} = 6 \times 10^{-10}$

$K_{spAgBr} = 3 \times 10^{-13}$

$\frac{2.303 \times R \times T}{F} = 0.0591$

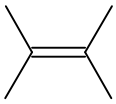
32.

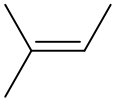
i)  $\xrightarrow[hv]{Cl_2}$ X (X = Number of Stereoisomers)

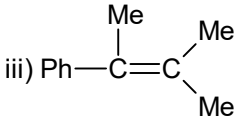
ii)  $\xrightarrow[hv]{Cl_2}$ Y (Y= Number of meso compound)

Find X + Y, considering monochlorination in each reaction.

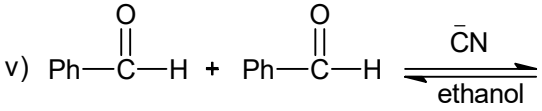
33. Total number of reactions in which product gives positive Tollen's test?

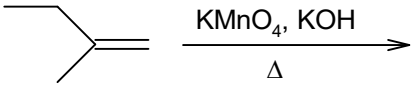
i)  $\xrightarrow[ii)\ DMS]{i)\ O_3}$

ii)  $\xrightarrow[ii)\ H_2O_2]{i)\ O_3}$

iii)  $\xrightarrow[ii)\ Zn,\ H_2O]{i)\ O_3}$

iv) $Ph-C\equiv C-Ph \xrightarrow[ii)\ H_2SO_4]{i)\ HgSO_4}$

v) 

vi) 

vii) Sucrose $\xrightarrow{H^+/H_2O}$

34. In how many reaction a gas (molecular weight = 28) is produced?

(i) $NH_4Cl \xrightarrow{\Delta}$

(ii) $(NH_4)_2SO_4 \xrightarrow{\Delta}$

(iii) $NH_4NO_2 \xrightarrow{\Delta}$

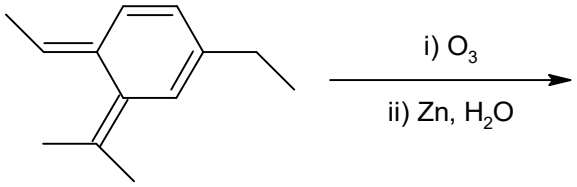
(iv) $NH_4NO_3 \xrightarrow[\text{above } 300^oC]{\text{Heating}}$

(v) $(NH_4)_2Cr_2O_7 \xrightarrow{\Delta}$

(vi) $CH_2CO \xrightarrow{\Delta}$

35. If number of plane of symmetry in $SiCl_4$ is x and ethene is y. Then find the value of $\frac{x^2 - y^2}{(x - y)}$.

36. Find the number of carbonyl compounds produced in below reaction which can respond to both haloform and Tollen's test.



Mathematics

PART – III

SECTION – A
(One Options Correct Type)

This section contains **06 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

37. The length of focal chord AB of ellipse $\frac{x^2}{4}+\frac{y^2}{3}=1$ is: $\left(\text{Given A}=\left(\frac{8}{5},\ \frac{3\sqrt{3}}{5}\right)\right)$

(A) $\frac{4}{5}$

(B) $\frac{16}{5}$

(C) $\frac{32}{5}$

(D) $\frac{64}{5}$

38. If $2x-y+1=0$ is a tangent on the parabola which intersect its directrix at (1, 3) and focus is (2, 1), then equation of axis of parabola is

(A) $11x-2y=24$

(B) $11x+2y+24=0$

(C) $2x-11y-24=0$

(D) $2y+11x=24$

39. If vectors $\hat{a}, \hat{b}, \hat{c}$ are in space such that $\hat{a}.\hat{b}=\hat{b}.\hat{c}=\hat{c}.\hat{a}=\frac{1}{2}$, then $\left(\hat{a}\times\hat{b}\right).\left(\hat{a}\times\left(\hat{b}\times\hat{c}\right)\right)$ is equal to

(A) $\frac{1}{\sqrt{2}}$

(B) $-\frac{1}{\sqrt{2}}$

(C) $\frac{1}{2\sqrt{2}}$

(D) $-\frac{1}{2\sqrt{2}}$

40. How many matrices X (3×3) with entries {0, 1, 2} are there for which sum of diagonal entries of $X.X^T$ is 7?

(A) 560

(B) 562

(C) 504

(D) 540

41. If α and β are roots of equation $\frac{3}{4}\sin\left(\frac{\theta}{9}\right) = \sin^3\theta + 3\sin^3\left(\frac{\theta}{3}\right) + 9\sin^3\left(\frac{\theta}{9}\right) + \frac{1}{4\sqrt{2}}$ for $0 < \theta < \frac{\pi}{2}$, then $\tan\alpha + \tan\beta$ is equal to

(A) $2+\sqrt{3}$

(B) $3+\sqrt{3}$

(C) $3-\sqrt{3}$

(D) $2-\sqrt{3}$

42. If $A = \left[a_{ij}\right]_{2\times 2}$, $B = \left[b_{ij}\right]_{2\times 2}$ such that $a_{ij} = f\left(x^{2i-j-2}\right)$ and $b_{ij} = f\left(x^{2i-j-1}\right)$ and

$$A + B^T = \begin{pmatrix} -x^2 f\left(x + f\left(1\right)\right) & \left(1-\frac{1}{x^4}\right) f\left(\frac{1}{x^2}\right) \\ f\left(\frac{1}{x}\right)\left(1-\frac{1}{x^2}\right) & -\frac{1}{x^2} f\left(\frac{1+f\left(1\right)}{x}\right) \end{pmatrix}$$, then $\int_x^{1/x} f\left(t\right) dt$ is always equal to

(A) $f\left(1\right)$

(B) $f\left(1\right)-1$

(C) $f\left(1\right)+1$

(D) None of these

(One or More than one correct type)

This section contains **06** multiple choice questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

43. If volume of a parallelepiped determined by the vectors $\bar{a}$, $\bar{b}$ & $\bar{c}$ is 2, then volume of parallelepiped determined by the vectors

(A) $3\left(\bar{a}+\bar{b}\right)$, $\left(\bar{b}+\bar{c}\right)$ and $\left(\bar{c}+\bar{a}\right)$ is 24

(B) $2\left(\bar{a}\times\bar{b}\right)$, $3\left(\bar{b}\times\bar{c}\right)$ and $\left(\bar{c}\times\bar{a}\right)$ is 24

(C) $\left(\bar{a}-2\bar{b}\right)$, $\left(\bar{b}-2\bar{c}\right)$ and $\left(\bar{c}-2\bar{a}\right)$ is 14

(D) $3\left(\bar{a}-\bar{b}\right)$, $\left(3\bar{b}-\bar{c}\right)$ and $\left(3\bar{c}-\bar{a}\right)$ is 52

44. If a point P is taken on xy = 2 and then a normal is drawn from P on the ellipse $\frac{x^2}{6}+\frac{y^2}{3}=1$ which is perpendicular to $x + y = 8$, then P is

(A) (1, 2)

(B) (–1, –2)

(C) (2, 1)

(D) (–2, –1)

45. A circle S having centre (α, β) intersect at three points A, B and C such that normals at A, B and C are concurrent at (9, 6) for parabola $y^2 = 4x$ and O is origin. Then,

(A) Sum of modulus of slopes of normals at points A, B and C is 6

(B) $\alpha+\beta = 4$

(C) Magnitude of normal having negative slope is 2

(D) Circle S also passes through O

46. If $f(x) = \left||x-1|+|x-3|-|2x-1|\right| + \left||x-1|+|x-3|+|2x-1|\right|$, then

(A) $f(x)$ is non-differentiable at x = 1

(B) $f(x)$ is non-differentiable at $x = \frac{3}{2}$

(C) $f(x)$ has minimum value is 2

(D) x = 1 is point of local minima

47. If $\int \cos 2018x \sin^{2016} x \, dx = f(x) + c$, where $f(0) = 0$, then

(A) $\int_0^{\pi} f(x)dx = f(\pi)$

(B) $f\left(\frac{\pi}{4}\right) = \frac{1}{2^{1009}} \cdot \frac{1}{2017}$

(C) $f\left(\frac{\pi}{4}\right) = \frac{1}{(2018)2^{1009}}$

(D) $f(x)$ is odd function

48. Let $f(x)$ be a continuous function which takes positive values for $x \geq 0$ and satisfy $\int_0^x f(t)dt = x\sqrt{f(x)}$ with $f(1) = \frac{1}{2}$. Then

(A) $f(x)$ is increasing function in domain

(B) $f(x)$ is decreasing function in domain

(C) $f\left(\sqrt{2}+1\right) = \frac{1}{2\sqrt{2}}$

(D) $f\left(\sqrt{2}+1\right) = \frac{1}{4}$

SECTION – C
(Numerical Answer Type)

This section contains **06** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

49. In a ΔABC, if $a^2 - b^2 - c^2 + (x-2)bc = 0$, then maximum integer value of x is …….

50. If $x + y = 3$ is tangent at (2, 1) on hyperbola, intersects asymptotes at A and B such that $AB = 8\sqrt{2}$. If centre of hyperbola is (–1, –1), then least possible value of sum of semi-transverse axis and semi-conjugate axis is $p\sqrt{q}$, where HCF (p, q) = 1 and $\sqrt{q}$ is irrational number. Then, $\left(\frac{p}{q}\right)$ is equal to

51. If $x^2 + y^2 = 16$ is director circle of locus of centre of a moving ellipse touches both the axes in first quadrant, then if major axis and minor axis are not fixed, then maximum value of product of semi-major and semi-minor axis is

52. If θ is acute angle of intersect between two curves $x^2 + 5y^2 = 6$ and $y^2 = x$, then $\tan\theta$ is equal to

53. If 8 harmonic means are inserted between two numbers a and b (a < b) such that arithmetic mean of a and b is $\frac{5}{4}$ times equal to geometric mean of a and b, then $\left(\frac{H_8 - a}{b - H_8}\right)$ is equal to

54. If $f(x) = \sum_{i=1}^{\infty} \frac{x}{\{(i-1)x+1\}(ix+1)}$, then $f(2017) + f\left(\frac{1}{2017}\right)$ is equal to